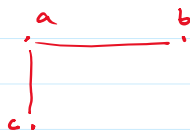


lecture 20.

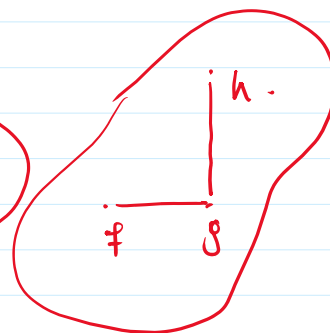
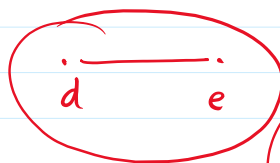
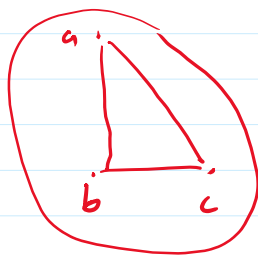
CONNECTIVITY

1- Connected Graph: Haibex if $\exists$ a path from "a" to "b".

p 563.

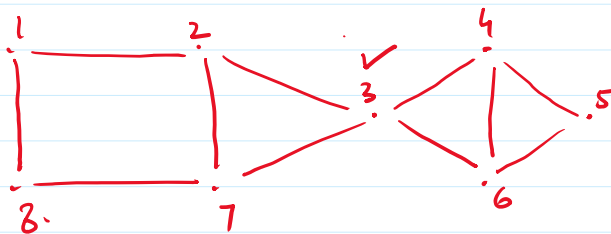


2 Connected Component:-

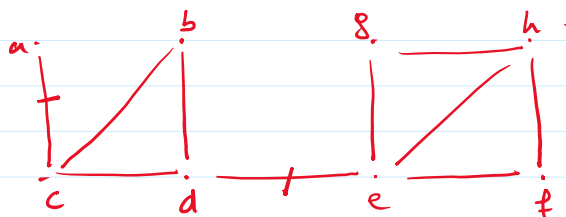


Cut Vertex:-

P564.



Cut Edge.



Cut Set.

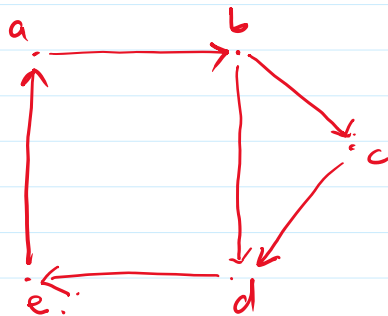
Self Study.

- Directed Graph.

Strongly Connected: $\forall a, b \in V. \exists$ a path from
 "a to b" & "b to a"

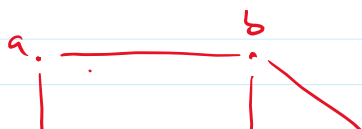
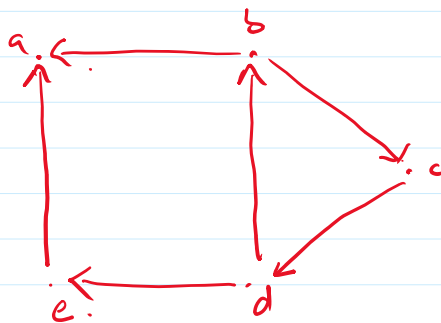
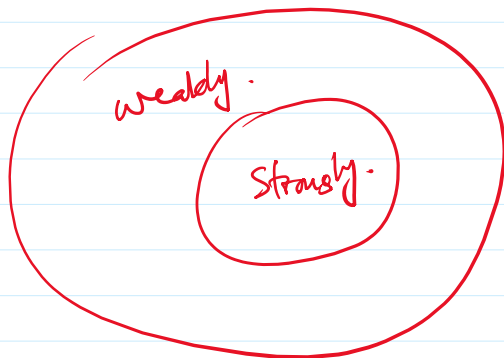
Weakly Connected: $\forall a, b \in V. \exists$ a path from
 "a to b" $\vee$ "b to a"

Ex: -
 S64

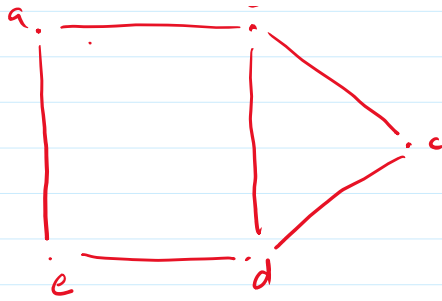


(a, b)

a to b	b to a.
a to c	c to a
a to d	d to a
a to e	e to a
b to c	c to b.
b to d	d to b
b to e	e to b.
c to d	d to c
c to e	e to c
d to e	e to d.



if the Underlying Undirected
 Graph is Connected.



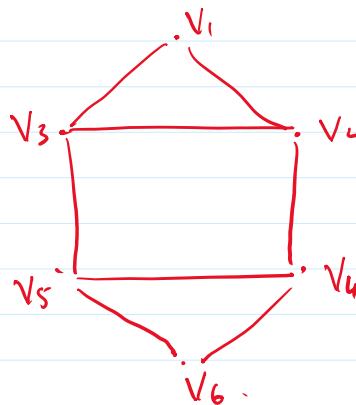
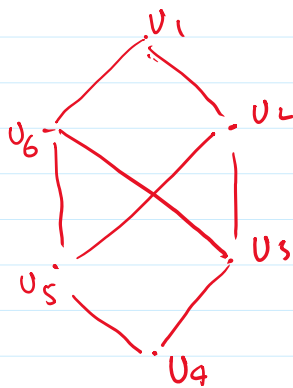
if the Underlying Undirected graph is Connected.
 $\rightarrow$ weakly Connected.

Isomorphism -

- 1)
- 2)
- 3)
- 4)
- 5) Cent Vertices -
- 6) Cent Edges -
- 7) Circuits -

Atit 9:00. Repet.

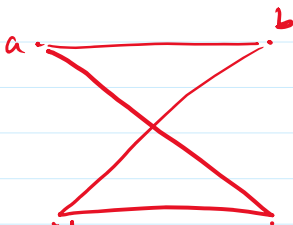
Ex 14 :-
 565



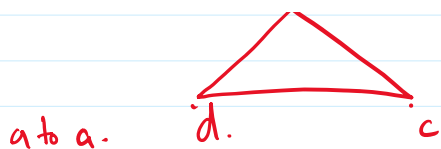
Length 3 (0 Circuits).

Length 5 (2 Circuits).

Ex 16
 567



$$A = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$



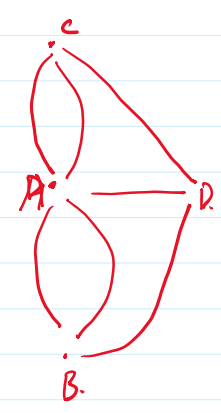
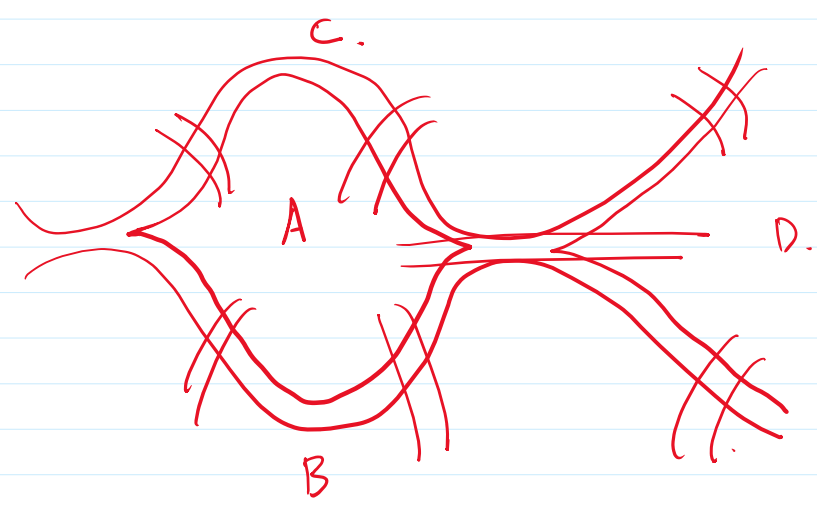
a b a b a a c a c a
 a b d b a a c a b a
 a b d c a a c d b a
 a b a c a a c d c a

$$d \begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{vmatrix}$$

$$A^4 = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 8 & 0 & 0 & 8 \\ 0 & 8 & 8 & 0 \\ 0 & 8 & 8 & 0 \\ 8 & 0 & 0 & 8 \end{bmatrix} \end{matrix}$$

EULER Circuit & PATH.

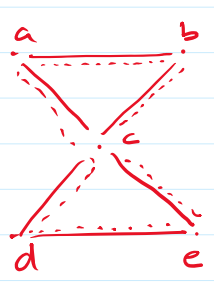
1736.



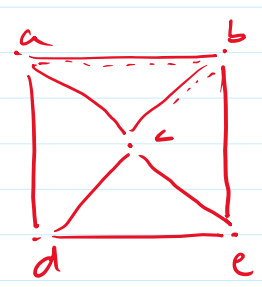
EULER PATH :- Traverse Each Edge Exactly once.

EULER Circuit :- $u \quad u \quad u \quad u \quad u \quad +$
Start = end.

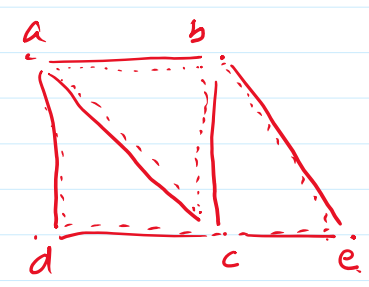
Ex2
P572



EP ✓
EC ✓.



EP X
EC X.

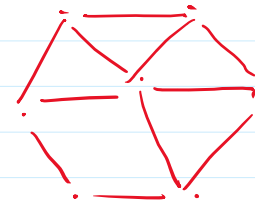
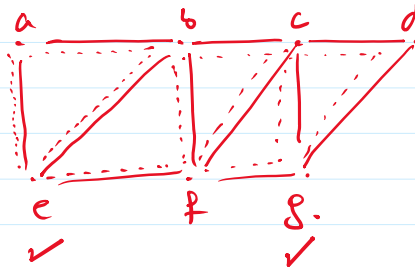
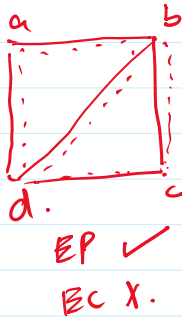


EP ✓
EC X.

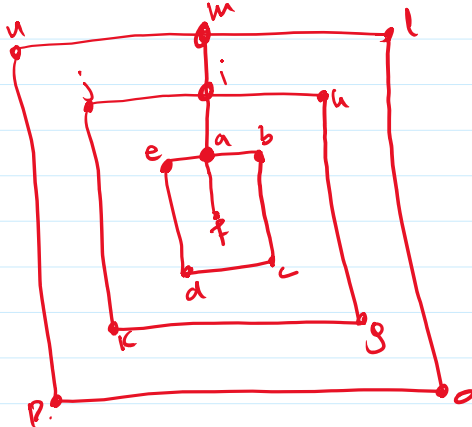
theorem 1:- A Connected Graph $|V| \geq 2$. has EC
 iff all Vertices has even degree.

theorem 2:- A Connected Graph. has EP but No EC
 iff it has Exactly 2 Vertices of odd degree.

Ex 9 :-
 575

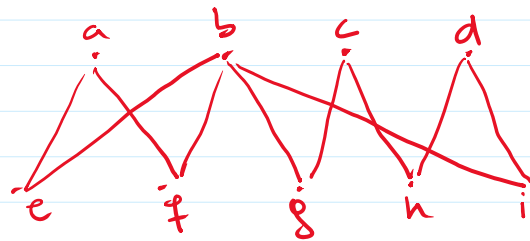


Quiz 7 13-11-2025



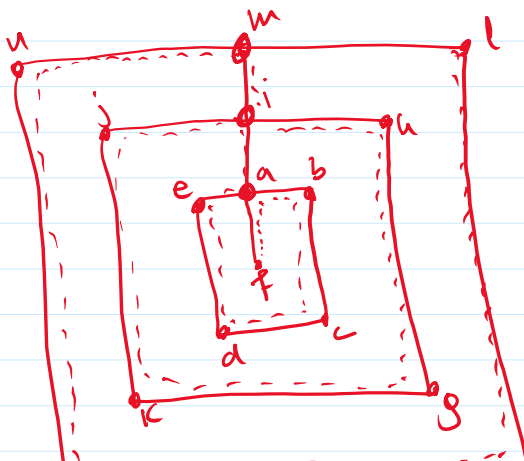
find EP & EC.

Quiz 8. 13-11-2025.

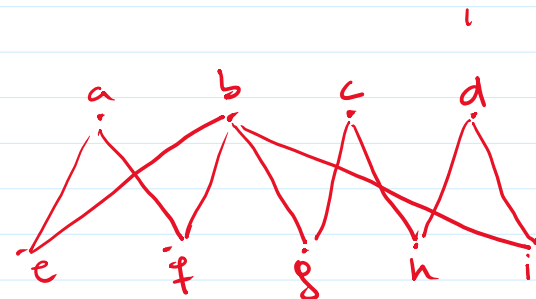


Write the Names of cut Vertices.

Solution:-



EC X.
 EP ✓.



Cut Vertices 'b'.

